

1. Administrative & Study Identification

Full Study Title: A Phase 1b, Open Label, Multi-center, Dose Optimization and Dose Expansion Study to Assess the Safety and Efficacy of DFV890 in Adult Patients With Myeloid Diseases **Short Title/Acronym:** Phase 1b Study of DFV890 in Adult Patients With Myeloid Diseases **Protocol Number:** CDFV890G12101 **NCT Number:** NCT05552469 **Sponsor Name:** Novartis Pharmaceuticals Corporation **Investigational Product:** DFV890 (Oral small molecule NLRP3 inhibitor) **Phase of Development:** Phase 1b **Medical Monitor:** Novartis Clinical Operations Lead / Medical Director

2. Study Synopsis & Rationale

2.1 Study Objective **Primary Objective:** To assess the safety, tolerability, pharmacokinetics, pharmacodynamics, efficacy, and recommended dose (RD) for single-agent DFV890. **Secondary Objectives:** To evaluate the preliminary efficacy of DFV890 based on hematologic improvement and Overall Response Rate (ORR), and further assess its pharmacokinetic profile (e.g., AUC, Cmax).

2.2 Background & Rationale Patients with lower-risk (LR) myelodysplastic syndromes (MDS) and chronic myelomonocytic leukemia (CMML) face shortened survival and high unmet medical need due to limited disease-modifying treatments, particularly after failing first-line agents. The NLRP3 pathway is abnormally activated in the bone marrow microenvironment of these myeloid neoplasms, driving smoldering inflammation, ineffective hematopoiesis, and clonal progression via IL-1 β and IL-18 production. DFV890, an NLRP3 inflammasome inhibitor, is hypothesized to block this pyroptotic cell death and cytokine secretion, thereby restoring normal hematopoiesis and providing disease-modifying clinical benefits.

3. Study Design & Methodology

3.1 Design Framework **Study Type:** Interventional **Control Type:** Single-arm (Dose-optimization and dose-expansion) **Blinding:** Open-label **Randomization:** Randomized (in the two-dose optimization part: low dose vs. high dose)

3.2 Planned Enrollment & Duration **Target N\$:** Approximately 105 adult patients **Number of Sites:** Multicenter, global (including USA, Singapore, Hong Kong, France, Germany, Italy) **Estimated Study Start:** 08-May-2023 **Estimated Primary Completion:** 27-Mar-2026

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Age ≥ 18 years at the time of signing the informed consent form (ICF). 2. Eastern Cooperative Oncology Group (ECOG) performance status ≤ 2 . 3. Diagnosis of IPSS-R defined very low, low, or intermediate-risk MDS who failed to respond to or did not tolerate ESAs, luspatercept, or HMAs; OR CPSS-defined LR CMML who failed/did not tolerate hydroxyurea or HMAs; OR High-Risk Clonal Cytopenia of Undetermined Significance (HR CCUS).

4.2 Exclusion Criteria 1. Systemic antineoplastic therapy or experimental therapy within 28 days or 5 half-lives prior to the first dose of study treatment. 2. Previous treatment with agents that have the same mechanism of action as DFV890 (e.g., targeting NLRP3 inflammasome or IL-1 pathway). 3. Receiving concomitant medications known to be strong/moderate modulators of cytochrome P450 enzymes CYP2C9 and/or CYP3A.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: DFV890 administered orally twice daily (BID). Patients are randomized to low or high dose during optimization, progressing to the Recommended Dose (RD) in the expansion phase. **Route of Administration:** Oral (tablets). **Duration of Treatment:** Minimum of 24 weeks (6 cycles of 28 days each) until disease progression, unacceptable toxicity, or patient/investigator decision to discontinue.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for managing MDS/CMML symptoms that does not conflict with CYP constraints. **Prohibited:** Hematopoietic colony-stimulating growth factors (e.g., G-CSF, GM-CSF), thrombopoietin mimetics, or erythroid-stimulating agents within 1 week prior to study treatment. Concomitant medications that strongly/moderately induce or inhibit CYP2C9 and/or CYP3A.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Baseline Bone Marrow Aspirate/Biopsy, Safety Labs
Baseline	Day 1	Randomization, First Dose, PK/PD blood sampling
Interim	Every 28 Days	Day 1 of each Cycle: Safety Labs, DLT Assessment, Efficacy evaluation, Drug accountability
End of Study	Follow-up	Final Vital Signs, Bone Marrow Assessment, AE/SAE resolution follow-up

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Assessment of safety, including the incidence and severity of Adverse Events (AEs), identification of Dose-Limiting Toxicities (DLTs), and determination of the optimal Recommended Dose (RD) for DFV890.

Measurement Method: Continuous monitoring using CTCAE criteria throughout the dose-evaluation periods.

7.2 Secondary & Exploratory Endpoints * Hematological improvement per 2006 IWG criteria for MDS and CMML. * Overall Response Rate (ORR) defined as Complete Remission (CR) or Partial Remission (PR). * Pharmacokinetics (PK) including AUClast and Cmax of DFV890.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Routine collection via site visits, patient reporting, and clinical laboratory checks throughout the 28-day cycles. **Serious Adverse Events (SAEs):** Standard reporting timelines to the sponsor within 24 hours of site awareness. **Data Monitoring Committee (DMC):** Safety Review Committee utilized to evaluate DLTs during the dose-optimization phase and declare the Recommended Dose (RD) for the expansion cohort.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Safety Set (all patients who receive ≥ 1 dose of DFV890); Efficacy Set (evaluable patients with baseline and post-treatment disease assessments). **Sample Size Justification:** $N \approx 105$ provides sufficient power for a Phase 1b trial to reliably estimate the maximum tolerated dose / recommended dose and to detect preliminary efficacy signals in expansion cohorts. **Handling of Missing Data:** Efficacy endpoints will rely on descriptive statistics and observed case analyses; missing safety data are generally not imputed.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.

Monitoring Plan: Scheduled onsite and remote monitoring visits by Novartis or designated CRO to ensure protocol adherence and patient safety. **Audit Trail:** Validated eCRF systems with comprehensive audit trails, utilizing data clarification forms for query resolution.

11. Study Identifiers & Governance

Primary ID: CDFV890G12101 **Registry Number:** NCT05552469 **Sponsor/Lead Organization:** Novartis Pharmaceuticals Corporation **Study Title:** Phase 1b Study of DFV890 in Adult Patients With Myeloid Diseases **Official Regulatory Title:** A Phase 1b, Open Label, Multi-center, Dose Optimization and Dose Expansion Study to Assess the Safety and Efficacy of DFV890 in Adult Patients With Myeloid Diseases

12. Clinical Framework (Myeloid Specific)

Primary Condition: Lower-risk Myelodysplastic Syndromes (LR MDS) and Chronic Myelomonocytic Leukemia (LR CMML) **Diagnosis Threshold:** IPSS-R defined very low, low, or intermediate risk MDS; CPSS-defined LR CMML **Medical Methodology:** Targeted biologic therapy via NLRP3 inflammasome inhibition **Optimization Target:** Disease modification via restoration of normal hematopoiesis and suppression of pyroptotic cell death

13. Study Procedures & Methodology

Management Pathway: Standard dose-finding (optimization) followed by single-arm dose expansion **Education Protocol (HCP):** Investigator protocol training, AE/DLT reporting guidelines, bone marrow biopsy standardization **Education Protocol (Patient):** Oral study drug adherence instruction and diary completion **Quality Audit Mechanism:** Independent central review of bone marrow responses and real-time safety monitoring

14. Observation & Follow-Up Schedule

Total Duration: Minimum 24 weeks of active treatment + survival/response follow-up **Onsite Visit Cadence:** Every 28 days (Day 1 of each cycle) **Remote Monitoring Frequency:** Routine risk-based data reviews **Checklist Review Interval:** End of each 28-day cycle

15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	_____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	_____	[DD-MMM-YYYY]				